Explainable AI Clinical Decision Support System

Clinical Prediction Report

Generated On: 05-09-2026 16:02:49

Patient Information

Field	Value
Patient ID	PAT003
Patient Name	John
Doctor	Dr. Smith
Hospital	Apollo Hospital
Disease	Stroke
Age	78
Gender	Male
Hypertension	Yes
Heart Disease	Yes
Ever Married	Yes
Work Type	Private
Residence Type	Urban
Average Glucose Level	245.60
Body Mass Index (BMI)	34.8
Smoking Status	never smoked

Prediction Result

Field	Result
Prediction	Stroke
Probability	94.34 %
Risk Level	High

Prediction Class	1

Clinical Recommendation

The model indicates a higher likelihood of stroke risk. Prompt consultation with a healthcare professional is strongly recommended for further clinical evaluation and appropriate diagnostic assessment.

Top Influencing Features (SHAP)

Feature	Impact
Age	2.7231
Average Glucose Level	0.6061
Hypertension	0.5017
Heart Disease	0.2217
Work Type - Children	-0.1588

LIME Explanation

Local Interpretable Model-Agnostic Explanation (LIME) showing the most influential conditions used by the model for this prediction.

Feature	Impact
age > 0.78	0.5502
work_type_children <= -0.40	-0.0999
hypertension > -0.33	0.0969
work_type_Self-employed <= -0.44	0.0627
avg_glucose_level > 0.17	0.0597

Generated by Explainable AI Clinical Decision Support System

Machine Learning • SHAP • LIME • Flask • React

Doctor Signature: _____

Patient Signature: _____